4.2.1 The Reversible RESPA Scheme

In many molecular dynamics applications, the system may be decomposed as $H(\boldsymbol{q},\boldsymbol{p}) = \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2 + U_{\mathrm{S}}(\boldsymbol{q}) + U_{\mathrm{F}}(\boldsymbol{q})$ where U_{S} is a "soft" potential, in which the natural motion would be relatively slow, whereas U_{F} is a much stronger term which would give rise to relatively rapid motion. We further suppose U_{S} (with its gradient) is expensive to compute, perhaps because it involves many terms, whereas U_{F} is relatively simple. Because U_{F} requires small timesteps, we think of separating the calculation of a timestep, typically in a symmetric form. Let $\mathscr{G}_h^{\mathrm{F}}$ denote the map associated to the Verlet method for the system without U_{S}, using a stepsize h:

$$\mathscr{G}_h^{\mathrm{F}} = e^{\frac{h}{2}\mathcal{L}_{U_{\mathrm{F}}}} e^{h\mathcal{L}_{\mathrm{K}}} e^{\frac{h}{2}\mathcal{L}_{U_{\mathrm{F}}}},$$

where $\mathcal{L}_{\mathrm{K}}$ is the Lie-derivative corresponding to the kinetic energy component of the Hamiltonian ($K = \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2$). Now consider the map constructed by taking r steps using $\mathscr{G}_{\mathrm{F},h/r}$ and composing this with kicks from the soft force:

$$\mathscr{G}_h^{\mathrm{MTS}} = e^{\frac{h}{2}\mathcal{L}_{U_{\mathrm{S}}}} [\mathscr{G}_{h/r}^{\mathrm{F}}]^r e^{\frac{h}{2}\mathcal{L}_{U_{\mathrm{S}}}}.$$

One step of the multiple timestepping algorithm is then implemented as follows:

Reversible RESPA Algorithm

$$\begin{aligned}
&\boldsymbol{P} := \boldsymbol{p} - (h/2)\nabla U_{\mathrm{S}}(\boldsymbol{q}),\\
&\boldsymbol{Q} := \boldsymbol{q},\\
&\quad \text{Repeat } r \text{ times:}\\
&\quad\left|\begin{array}{l}
\boldsymbol{P} := \boldsymbol{P} - (h/2)\nabla U_{\mathrm{F}}(\boldsymbol{Q})/r,\\
\boldsymbol{Q} := \boldsymbol{Q} + h\boldsymbol{M}^{-1}\boldsymbol{P}/r,\\
\boldsymbol{P} := \boldsymbol{P} - (h/2)\nabla U_{\mathrm{F}}(\boldsymbol{Q})/r,
\end{array}\right.\\
&\boldsymbol{P} := \boldsymbol{P} - (h/2)\nabla U_{\mathrm{S}}(\boldsymbol{Q}).
\end{aligned} \tag{4.2}$$

Multiple timestepping is widely used for efficiency purposes in systems with expensive force computations, but it should be used cautiously. To illustrate the potential dangers of multiple timestepping, one need only consider its application to a linear one-DOF model problem

$$H(q,p) = p^2/2 + U_{\mathrm{S}}(q) + U_{\mathrm{F}}(q), \qquad U_{\mathrm{S}}(q) = q^2/2, \qquad U_{\mathrm{F}}(q) = \Omega^2 q^2/2,$$